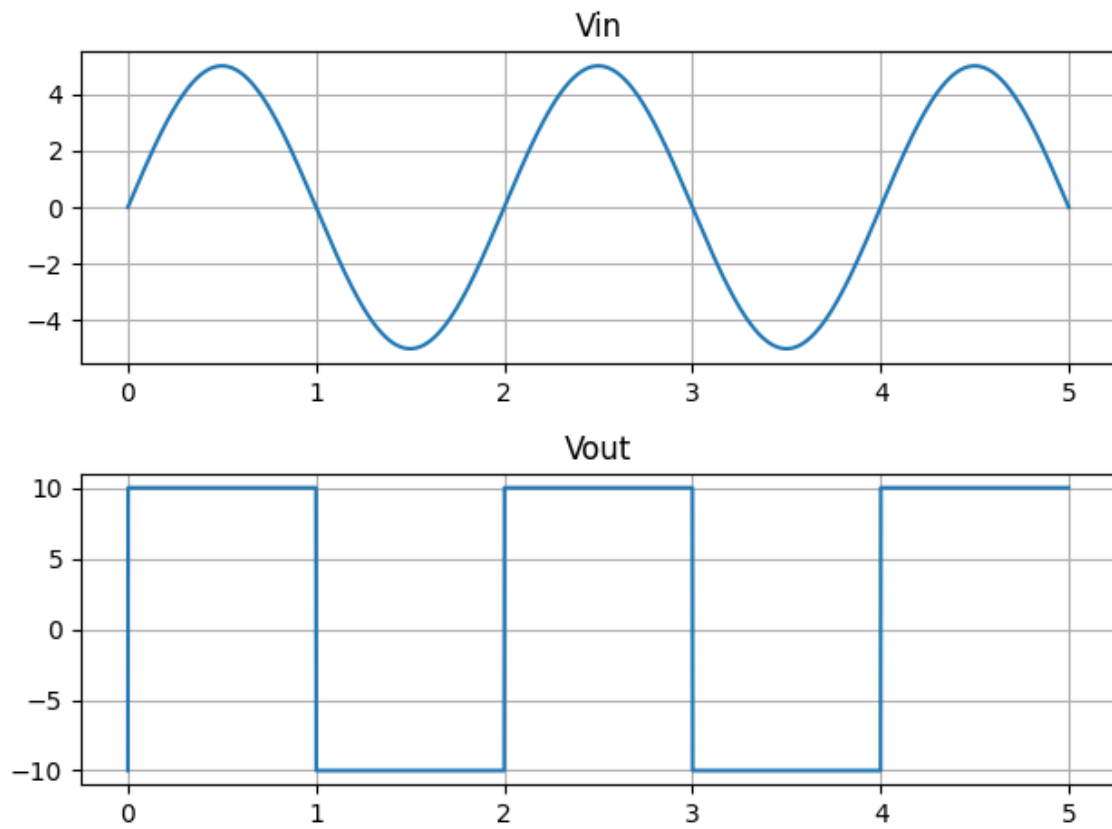


Exercise 2

Solution:

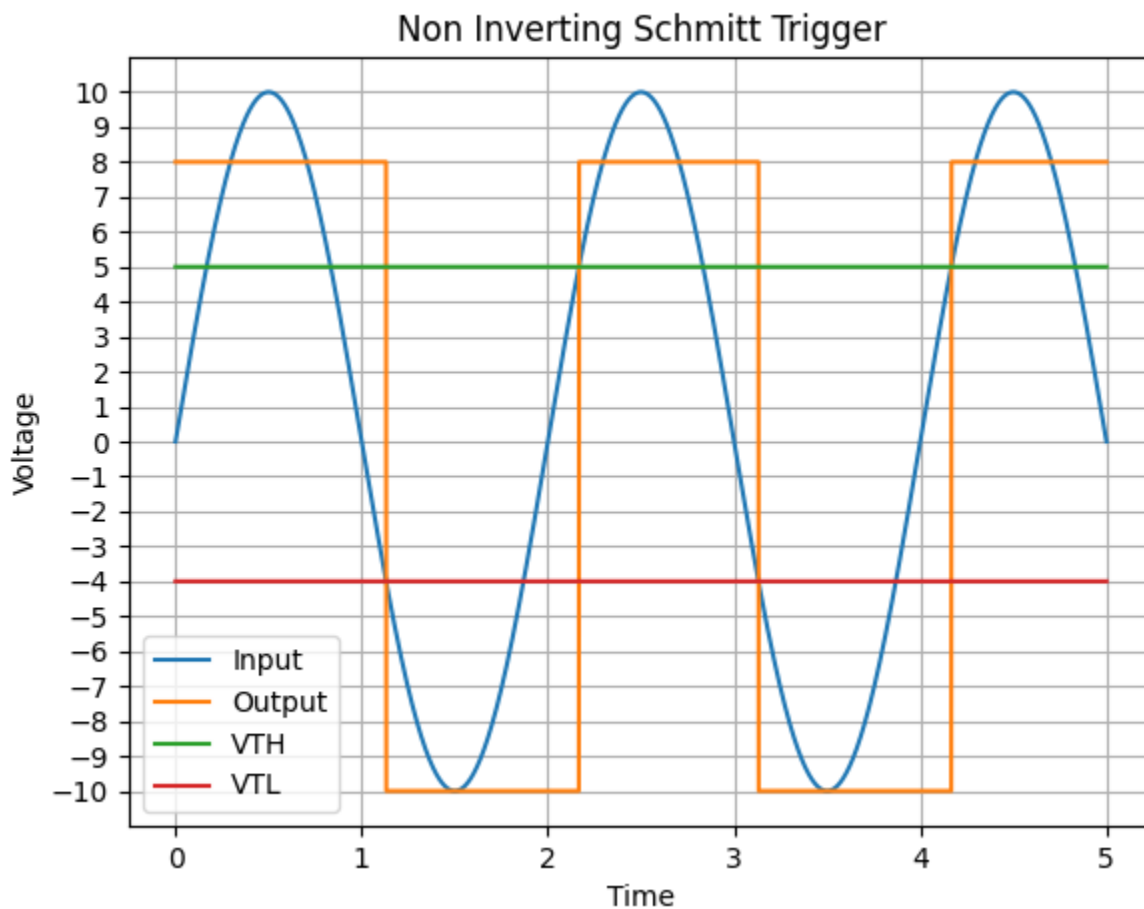


Exercise 3

Solution:

$$V_{TH} = -\frac{R_1}{R_3} \times V_L = -\frac{10}{20} \times -10 = 5 \text{ V}$$

$$V_{TL} = -\frac{R_1}{R_3} \times V_L = -\frac{10}{20} \times 8 = -4 \text{ V}$$



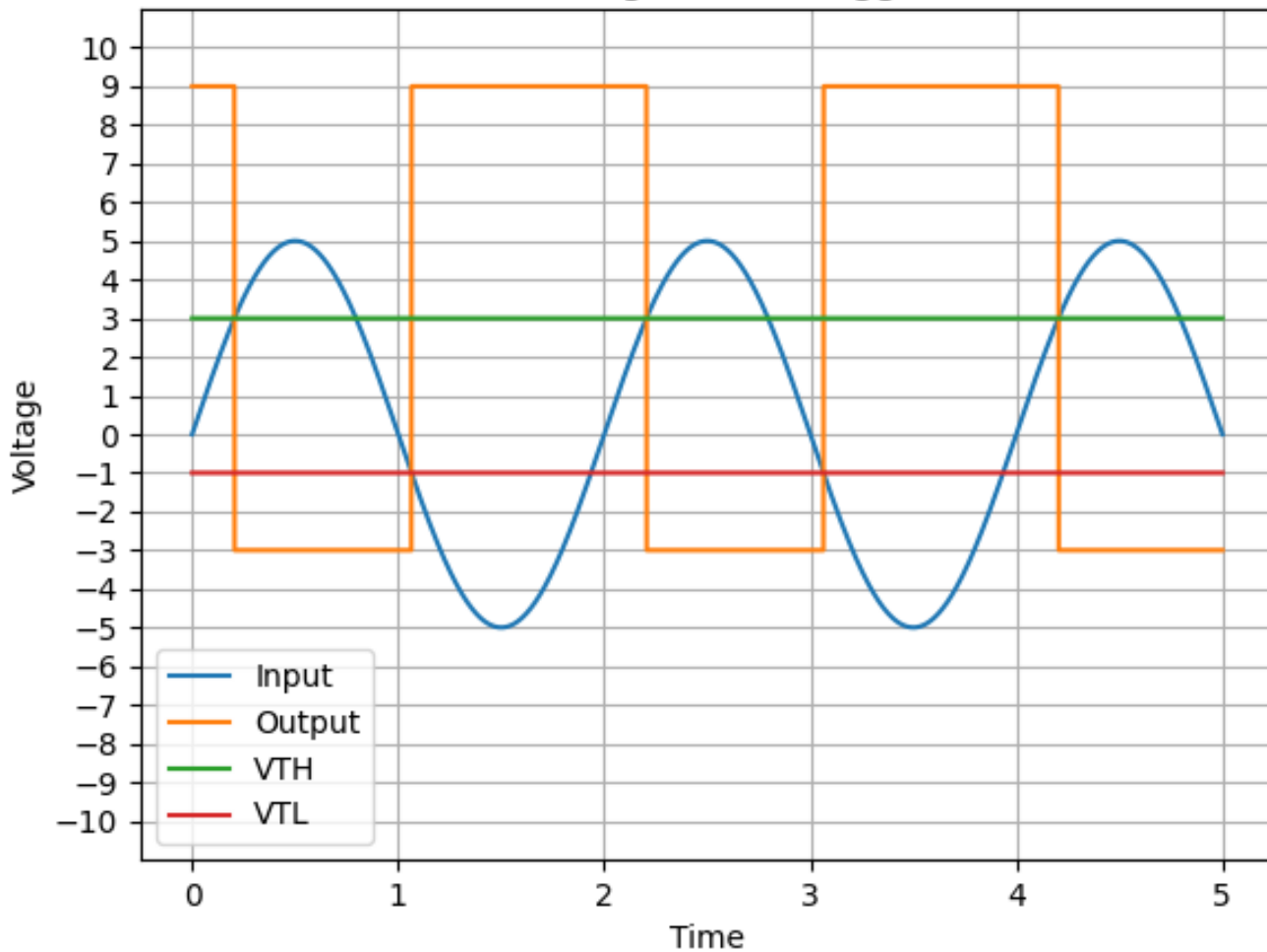
Exercise 4

Solution:

$$V_{TH} = \frac{R_1}{R_1 + R_3} \times V_H = \frac{10}{10 + 20} \times 9 = 3 \text{ V}$$

$$V_{TL} = \frac{R_1}{R_1 + R_3} \times V_L = \frac{10}{10 + 20} \times -3 = -1 \text{ V}$$

Inverting Schmitt Trigger



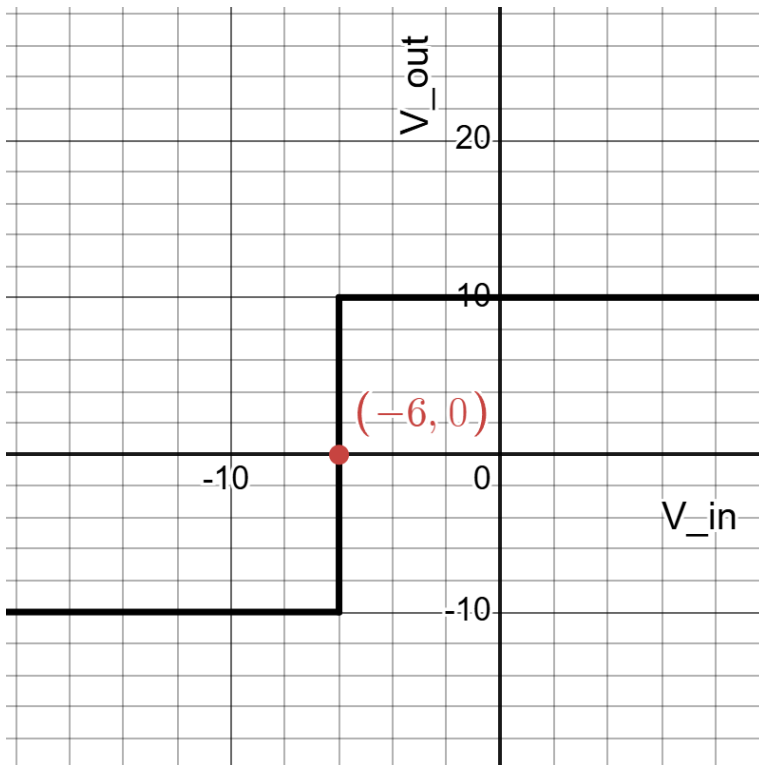
Exercise 5

Solution:

For input & reference voltage both applied to non-inverting terminal.

$$\text{If, } V_{in} > -\frac{R_2}{R_1} \times V_{ref} = -\frac{20}{10} \times 3 = -6 \text{ V, then } V_o = V_H$$

$$\text{If, } V_{in} < -\frac{R_2}{R_1} \times V_{ref} = -6 \text{ V, then } V_o = V_L$$



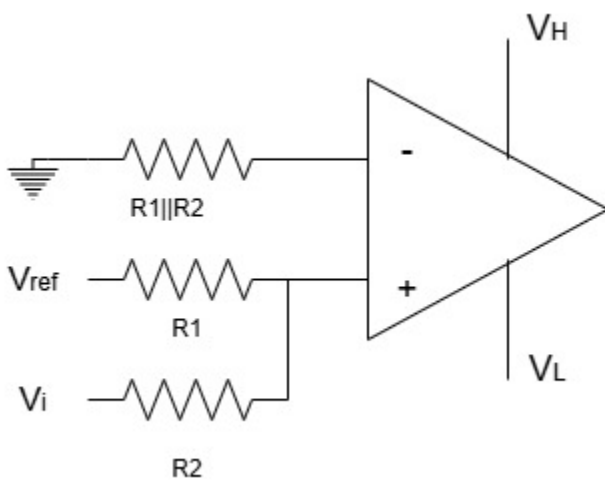
Exercise 6

Solution:

Assuming $R_1 = 5\text{ k}\Omega$ & $R_2 = 10\text{ k}\Omega$,

$$-\frac{R_2}{R_1} \times V_{ref} = 8 \rightarrow V_{ref} = -8 \times \frac{R_1}{R_2} = -8 \times \frac{5}{10} = -4\text{ V}$$

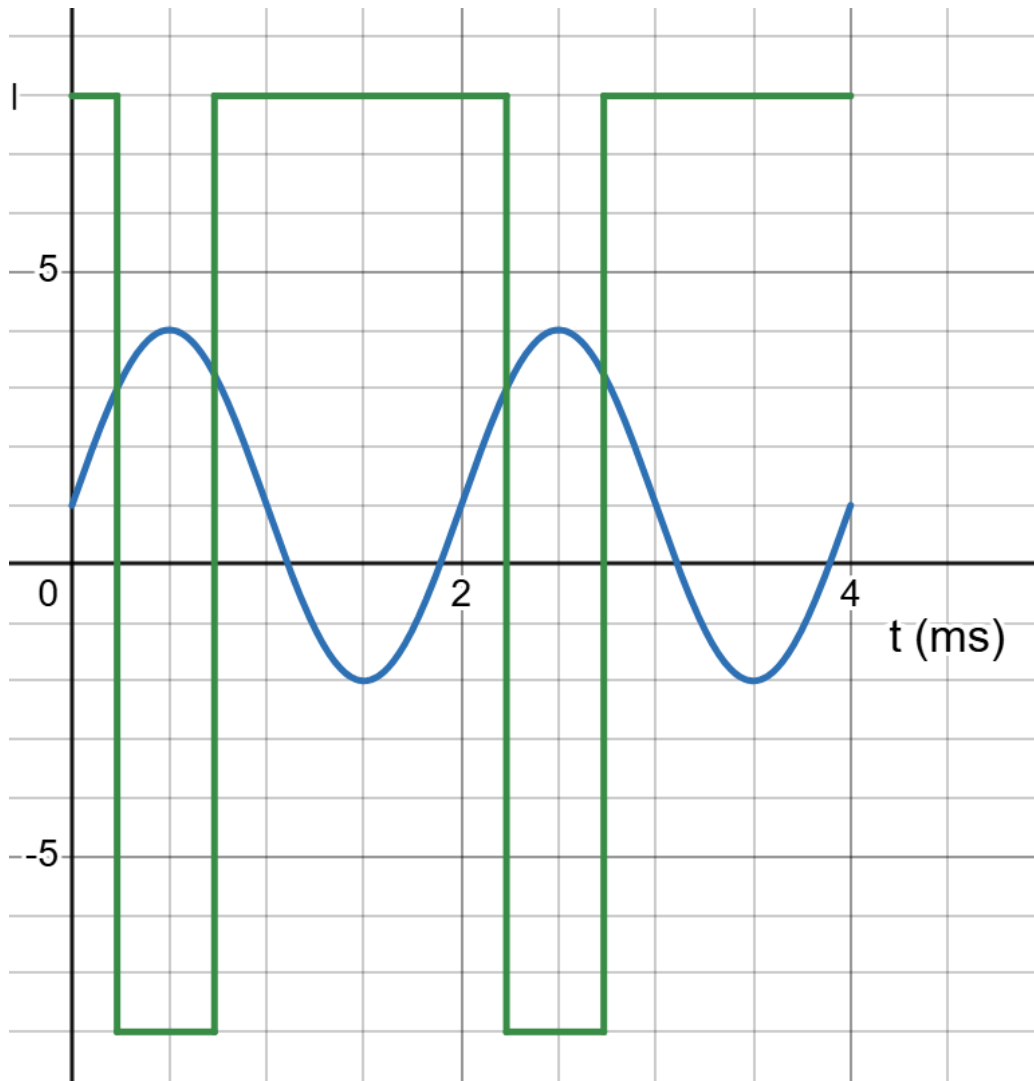
$$V_H = 10\text{ V}, V_L = -10\text{ V}$$



Exercise 7

Solution:

$$V_i < -\frac{R_2}{R_1} \times V_{ref} = -\frac{15}{10} \times -2 = 3 \text{ V}$$



The green graph is V_{out} vs time.

Exercise 8

Solution:

Non-Inverting:

$$V_{TH} = \frac{R_1+R_2}{R_2} \times V_{ref} - \frac{R_1}{R_2} \times V_L = \frac{5+20}{20} \times 3 - \frac{5}{20} \times -10 = 6.25 \text{ V}$$

$$V_{TL} = \frac{R_1+R_2}{R_2} \times V_{ref} - \frac{R_1}{R_2} \times V_H = \frac{5+20}{20} \times 3 - \frac{5}{20} \times 12 = 0.75 \text{ V}$$

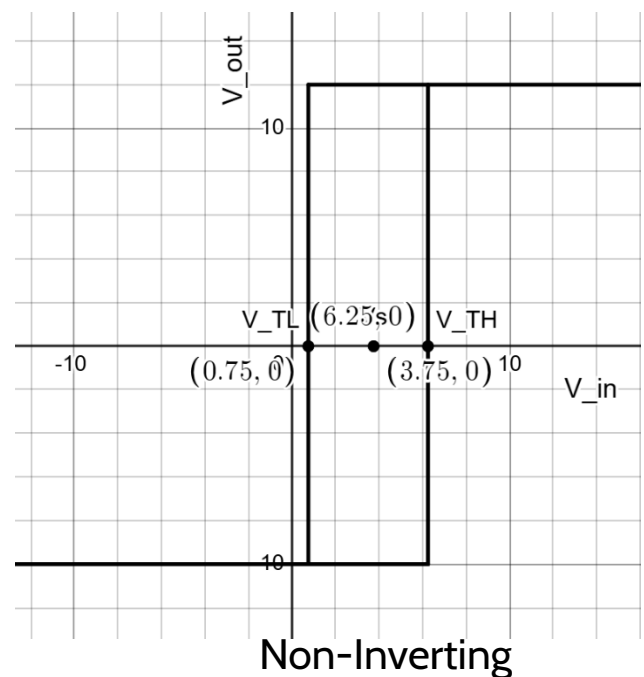
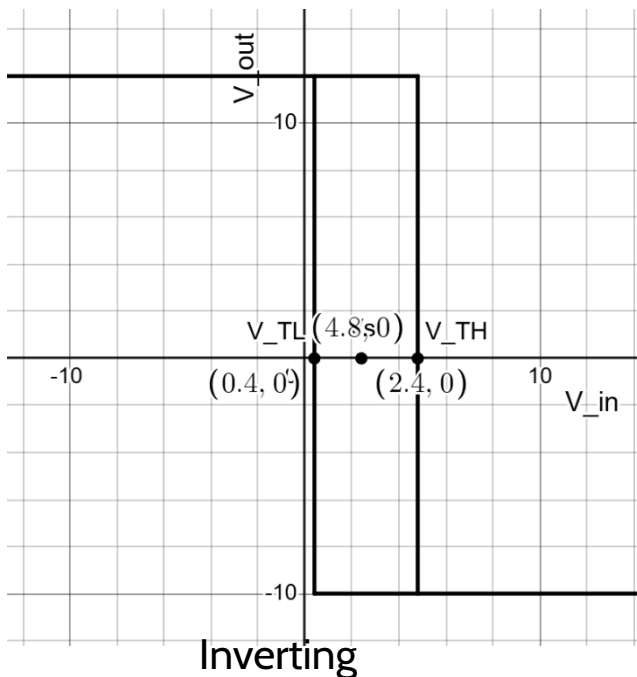
$$V_S = \frac{R_1+R_2}{R_2} V_{ref} = \frac{5+20}{5} \times 3 = 3.75 \text{ V}$$

Inverting:

$$V_{TH} = \frac{R_2}{R_1+R_2} \times V_{ref} + \frac{R_1}{R_1+R_2} \times V_H = \frac{5}{5+20} \times 3 + \frac{5}{5+20} \times 12 = 4.8 \text{ V}$$

$$V_{TL} = \frac{R_2}{R_1+R_2} \times V_{ref} + \frac{R_1}{R_1+R_2} \times V_L = \frac{5}{5+20} \times 3 + \frac{5}{5+20} \times -10 = 0.4 \text{ V}$$

$$V_S = \frac{R_2}{R_1+R_2} = 2.4 \text{ V}$$



Exercise 9

Solution:

The VTC is of an inverting Schmitt Trigger.

$$V_S = 0.8 = \frac{R_1}{R_1 + R_2} \times V_{ref} = \frac{2}{R_2 + 2} \times V_{ref} \rightarrow V_{ref} = \frac{0.8(2 + R_2)}{2}$$

$$V_{TH} = V_S + \frac{R_1}{R_1 + R_2} \times V_H \rightarrow 4 = 0.8 + \frac{2}{R_2 + 2} \times 10 \rightarrow \frac{2}{R_2 + 2} = \frac{3.2}{10} \rightarrow R_2 = 4.25 \text{ k}\Omega$$

$$\therefore V_{ref} = 2.5 \text{ V}$$